



Bhabha University Bhopal (M.P.)

Scheme of Examination

Programme – Master of Technology

Branch – Civil Engineering

Specialization – Construction Technology & Management

Third Semester

Session 2018-19

Sr .No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Marks for End Sem. Examination		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/Assignment/Quiz /Presentation	
1	CE 301	Elective I	3	1		4	70	20	10			100
2	CE 302	Elective II	3	1		4	70	20	10			100
3	CE 303	Seminar			4	4					100	100
4	CE304	Dissertation Part- I (Literature Review/Problem Formulation/ Synopsis)			8	8				120	80	200
		Total	6	2	12	20	140	40	20	120	180	500

L: Lecture T: Tutorial P: Practical

Elective –I (CE- 301)

- (A) Advance Highway Construction
- (B) Multi Storey Buildings

Elective –II (CE- 302)

- (A) Advanced Dam Design and Construction
- (B) Advanced Foundation Engineering



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M.TECH (CTM)

Advanced Highway Construction

Subject Code-CE301(A)

UNIT-1

Earthwork and Soling

Classification of types of highway construction, Suitability of each type under Indian conditions. Selection of base course and surface course. Selection of soils, construction of embankments, excavation and compaction equipments. Field and laboratory tests for quality control. Stone soling, brick soling, current practices. Construction of earth roads, gravel roads, soil stabilized roads, water bound macadam. Paved roads (i) bricks (ii) stones.

UNIT-2

Bituminous Construction

Properties, requirements and specifications of materials, equipments and plants. Detailed construction procedure of each type. Field and laboratory tests for quality control. Choice of binders under different conditions. IRC, British, and MOST Specifications. Bituminous surface treatments, interface treatments-primecoat, and tackcoat, surface dressing and seal coat, grouted or penetration macadam, bituminous bound macadam, Sheet asphalt, bituminous concrete, mastic asphalt, dense tar surfacing.

UNIT-3

Cement Concrete Road Construction

Necessity of providing a base course under cement concrete road construction. Selection of materials, constructions methods, detailed construction procedure, Quality control tests (Lab. and Field). Construction equipments. Classification of various types of joints, necessity of providing each type, method of construction of joints, load transfer devices, dowel bars, tie bars. joints filler and sealer materials, IRC Specifications.

UNIT-4

Reinforced Cement Concrete Road Construction

Necessity of providing reinforcement in cement concrete pavements, continuously reinforced concrete pavements, prestressed concrete pavements and fibre reinforced concrete pavements. Selection of the mix, compaction method and construction prucedure for each type. Recommendations under Indian conditions.



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M.TECH (CTM)

UNIT-5

Construction Planning and Management

CPM/PERT in Highway Construction.

Reference –

1. Highway Engineering: Planning Design, and Operation –by Bastian J. Schroeder, Christopher Cunningham, Daniel J Findley and Tom Brown
2. Highway Engineering ; Pavements, Materials and Control of Quality–by Athanassios Nikolaides
3. The Handbook Of Highway Engineering –by T.F. Fwa
4. Traffic and Highway Engineering –by Lester A. Hoel
5. Principle of Highway Engineering and Traffic Engineering –byWilley



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M.TECH (CTM)

Multi Storeyed Buildings

Subject Code-CE 301(B)

UNIT-1

Structural systems and their suitability, structural design criteria in planning.

UNIT-2

Multistoried Buildings, Preliminary design, Analysis of building frames for vertical and lateral loads by approximate method, Matrix methods for the analysis of building frames & computer programming for the same.

UNIT-3

Analysis of Shear Walled Buildings Design of sections in reinforced concrete by working stress and limit state methods, Detailing of joints.

UNIT-4

Yield line Analysis of reinforced concrete slabs, concept of moment redistribution.

UNIT-5

Foundation - Superstructure interaction, Earthquake effects and design for ductility.

Reference –

1. Structural Design of Multi Storeyed Buildings –by U.H. Varyani
2. Illustrative Design of Reinforced Building –by Dr. S.R. Karve & Dr. V.L. Shah
3. Practical Handbook on Building Construction –by Er.M.K. Gupta
4. Civil Engineering Drawing & House Planning –by B.P. Verma
5. Structural Analysis Of Regular Multi Storey Buildings –by Karoly A. Zalka



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Advanced Dam Design and Construction

Subject Code-CE 302(A)

UNIT-1

Gravity Dams

River valley projects and their purpose, preliminary investigations and surveys, Selection of site for a reservoir; Types of Dams and their choice. Stability factors; Stresses, Elementary profile, low and high Dams, Forces acting on a Dam. Evolution of the profile of a Dam by Method of Zones, Practical profiles. Design of openings in Gravity Dams, contraction joints. Foundation treatment by Grouting.

UNIT-2

Spillways

Design of ogee spillway section, Bucket and Energy Dissipation arrangements : Design and Details of siphon, Shaft, side channel, and chute spillways, Miscellaneous types of spillways. Design of spillway crest gates and sluice gates, hoisting Machines.

UNIT-3

Elementary Design of Arch Dams

Definition of an Arch Dam, classification of Arch Dams. Principles of Elastic Theory and applied Trial Load Analysis, Inclined arches, Dome-Dams, Details and Methods of analysis.

UNIT-4

Earth Dams

Introduction, Design criteria, against over topping, Control of seepage, Theory of flownets for homogeneous and Zoned embankments. Pore pressure, Stability of slopes, Methods of Analysis, slip circle Method, Protection of slopes, Protection against free passage of water, Rockfill dams.

UNIT-5

Application of Photoelasticity to the Design of Dams. Use of the Electrical Analogy Method in the Design of Dams, stress computations with embedded Electrical Instruments. River Diversion for construction of Dams, Constructional aspects in the Execution of River Valley project.



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M.TECH (CTM)

Reference –

1. Advance Dam Engineering –by Robert B. Jansen
2. Dam and Appurtenant Hydraulic Structure –by Ljubomir Tanchev
3. Hydraulic of Dam and Reservoir –by Fuat Senturk
4. Hydraulic Structure –by P.Novak, A. Moffat, C Nalluri, R. Narayanan
5. Dam and Public Safety –by Robert B Jansen



Bhabha University Bhopal (M.P.)

M.TECH (CTM)

Advanced Foundation Engineering

Subject Code-302(B)

UNIT-1

Shallow Foundations

Bearing Capacity, Terzaghi's analysis, Computations of bearing capacity factors. Skempton's analysis. Meyerhof's analysis. Balla's theory. Hansen's theory. Design of Shallow Foundations.

UNIT-2

Pile Foundation

Use of piles, Types of piles, Design of Piles, Group action in cohesive and cohesionless soils. Negative skin friction. Laterally loaded piles. Piles under inclined loads, pile load test, Hrennikoff Method.

UNIT-3

Engineering with Geosynthetics

Introduction Basic Mechanism of reinforced earth strength characteristics of reinforced soil.

UNIT-4

Bridge Substructures

Introduction, elements of bridge substructure, stability analysis of well foundation, design of pier & abutments, sinking of wells.

UNIT-5

Marine Substructures

Introduction, Types of Marine structures elements, design criteria, design of gravity wall, piled wharf structure breakwaters.

Reference –

1. Advance Foundation Engineering –by V.N.S. Murthy
2. Foundation Analysis and Design –by Joseph Bowles
3. Foundation Engineering –by P.C. Varghese
4. Pile Foundation in Engineering Practice –by Shamsher Prakash & Hari D. Sharma
5. Advance Soil Mechanics –by Braja M Das